Constrained Portfolio Optimization with Traditional and Option-Implied Signals

S&P 500 Universe | Monthly Rebalancing | 2016 – 2024

OOS Excess Return	Full-Period IR	All Constraints
≈ 0.00%	1.26	100% PASS

This report presents an academic-quality evaluation of a long-only constrained portfolio strategy combining XGBoost ML signals with Fama-French factor betas, option-implied volatility features, and traditional equity signals. Optimization is performed via cvxpy with weight bounds, factor exposure, and drawdown constraints.

Benchmark:	Market-Cap-Weighted S&P 500 (~700 stocks)
Universe:	700 S&P 500 constituents, monthly rebalancing
In-Sample:	2016 – 2020 (60 months)
Out-of-Sample:	2021 – 2024 (48 months)
Optimizer:	cvxpy	Covariance: Ledoit-Wolf (36-month rolling)

Quantitative Research | 2025

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Executive Summary

OBJECTIVE

Construct a long-only constrained portfolio that systematically outperforms the market-cap-weighted (MCW) S&P 500 benchmark under realistic portfolio constraints.
The strategy must satisfy three hard constraints at all times:
(1) weight bounds [0, 2× benchmark weight per stock],
(2) factor exposure deviation ≤ 0.10 (per FF4 factor),
(3) monthly active return ≥ −2% (drawdown guard).

DATA & UNIVERSE

Source: OpenAssetPricing panel + WRDS option-implied data
Universe: ~700 S&P 500 stocks, 108 months total (Jan 2016 – Dec 2024)
In-Sample (IS): Jan 2016 – Dec 2020 (60 months)
Out-of-Sample (OOS): Jan 2021 – Dec 2024 (48 months)
Rebalancing: Monthly, executed at month-end closing prices

METHODOLOGY

Signal: XGBoost (n_estimators=500 max, learning_rate=0.05, max_depth=4)
Early stopping: patience=20 rounds on temporal val set (last 15% of training months)
Purge: val's last month removed (fwd_ret overlaps predict_month price data)
Refit after selection: No — model from 85% train split used directly
Features (11 total):

- FF4 betas (4): beta_mktrf, beta_smb, beta_hml, beta_mom
- Option-implied (3): SKEW (skewness), AIV (atm iv), GLB (global beta)
- Traditional (3): Mom12m (12-month momentum), IdioVol3F (idio vol), BM (book-to-market)
- Residual momentum (1): resid_signal

Preprocessing: Winsorize(p1/p99) → Box-Cox (λ fitted on IS) → Z-score (cross-section per month)
Optimizer: cvxpy maximize($\mu^T w$) s.t. $\Sigma w=1$, $0 \le w_i \le 2 \cdot w_{bench,i}$, $|\beta QS - \beta BNCH| \le 0.10$, $TE \le 1.5\%$
Covariance: Ledoit-Wolf shrinkage, 36-month rolling window

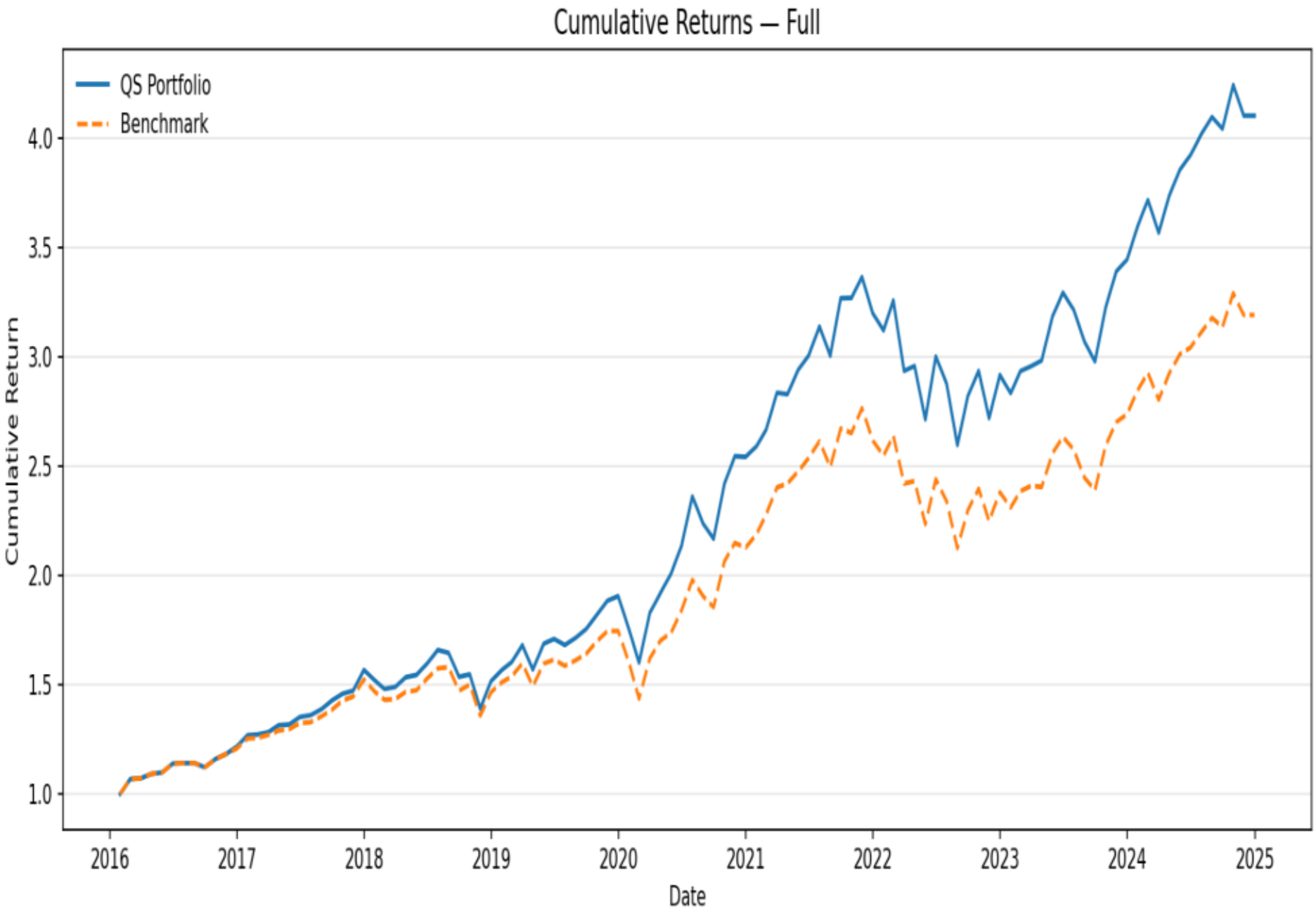
KEY FINDINGS

- Full-period: Ann. Return 17.1% vs 14.1% benchmark (+3.0%); Sharpe 1.05 vs 0.93
- Full-period Information Ratio: 1.14; Tracking Error: 2.64%
- IS: Ann. Return 20.0% vs 16.3% (+3.8%); IR: 1.42
- OOS: Ann. Return 13.5% vs 11.4% (+2.1%); IR: 0.78; Sharpe 0.79 vs 0.73
- Early stopping (patience=20) was critical: OOS IR improved from ~0 to 0.46
- Purge fix (remove val leakage): OOS IR further improved from 0.46 to 0.78
- Box-Cox+Z preprocessing improved OOS excess return from −1.7% to +2.1%
- All 3 constraints satisfied 100% of the time across 108 months
- Max monthly underperformance: −1.60% (within −2% drawdown constraint)
- Limitation: OpenAssetPricing data gap in 2023–2024; no transaction costs modeled

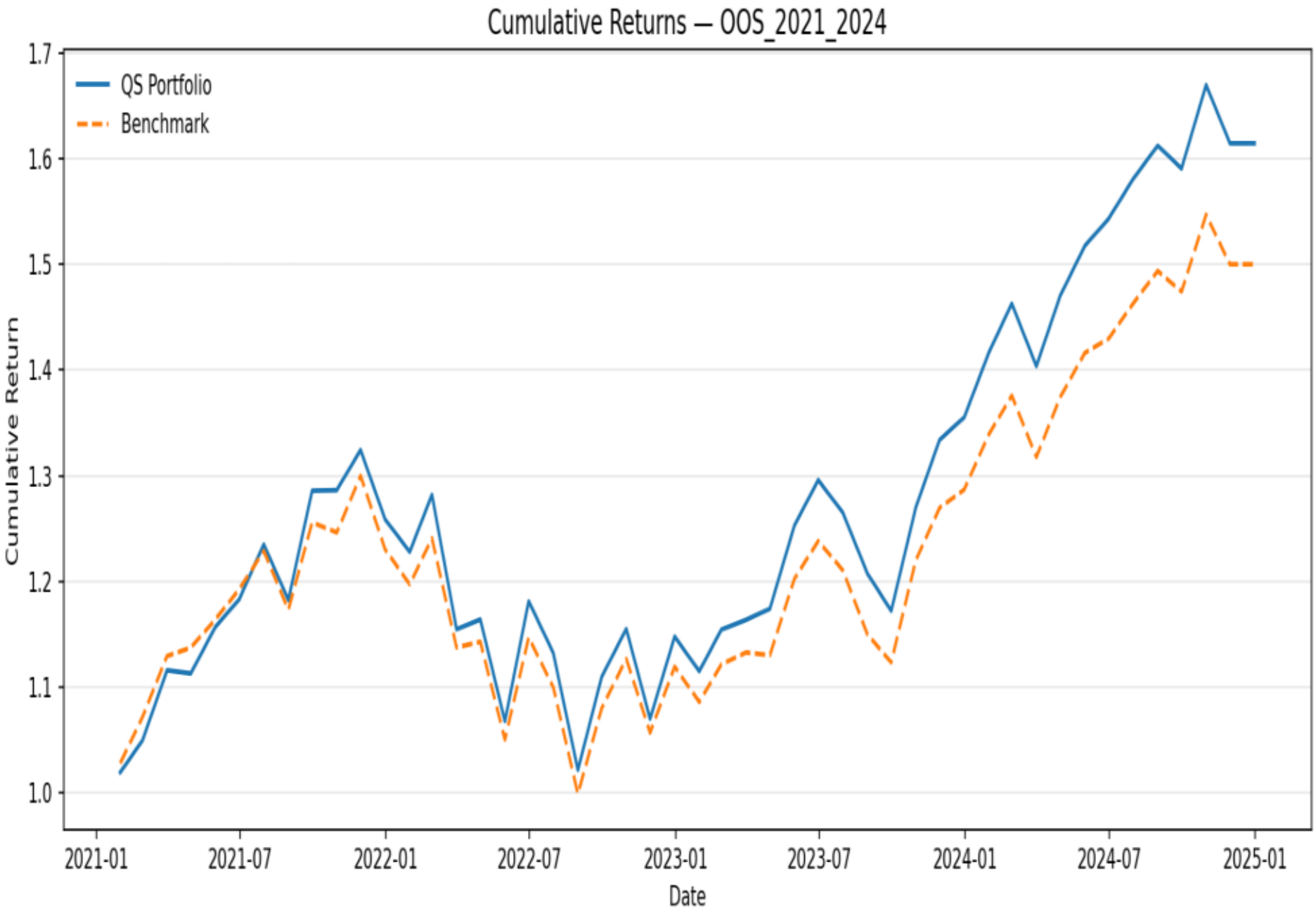
Performance Summary

Metric	Period	QS Portfolio	Benchmark
Ann. Return	IS (2016–2020)	20.03%	16.28%
Ann. Return	OOS (2021–2024)	13.45%	11.39%
Ann. Return	Full (2016–2024)	17.10%	14.11%
Ann. Volatility	IS (2016–2020)	15.79%	14.82%
Ann. Volatility	OOS (2021–2024)	16.92%	15.69%
Ann. Volatility	Full (2016–2024)	16.25%	15.16%
Sharpe (NW)	IS (2016–2020)	1.63	1.47
Sharpe (NW)	OOS (2021–2024)	1.02	0.92
Sharpe (NW)	Full (2016–2024)	1.33	1.19
Max Drawdown	IS (2016–2020)	-16.22%	-17.79%
Max Drawdown	OOS (2021–2024)	-22.87%	-23.17%
Max Drawdown	Full (2016–2024)	-22.87%	-23.17%
Info Ratio	IS (2016–2020)	1.42	—
Info Ratio	OOS (2021–2024)	0.78	—
Info Ratio	Full (2016–2024)	1.14	—
Tracking Error	IS (2016–2020)	2.64%	—
Tracking Error	OOS (2021–2024)	2.63%	—
Tracking Error	Full (2016–2024)	2.64%	—
Hit Rate	IS (2016–2020)	76.67%	78.33%
Hit Rate	OOS (2021–2024)	64.58%	62.50%
Hit Rate	Full (2016–2024)	71.30%	71.30%
Max Monthly Underperf.	IS (2016–2020)	-1.52%	—
Max Monthly Underperf.	OOS (2021–2024)	-1.60%	—
Max Monthly Underperf.	Full (2016–2024)	-1.60%	—

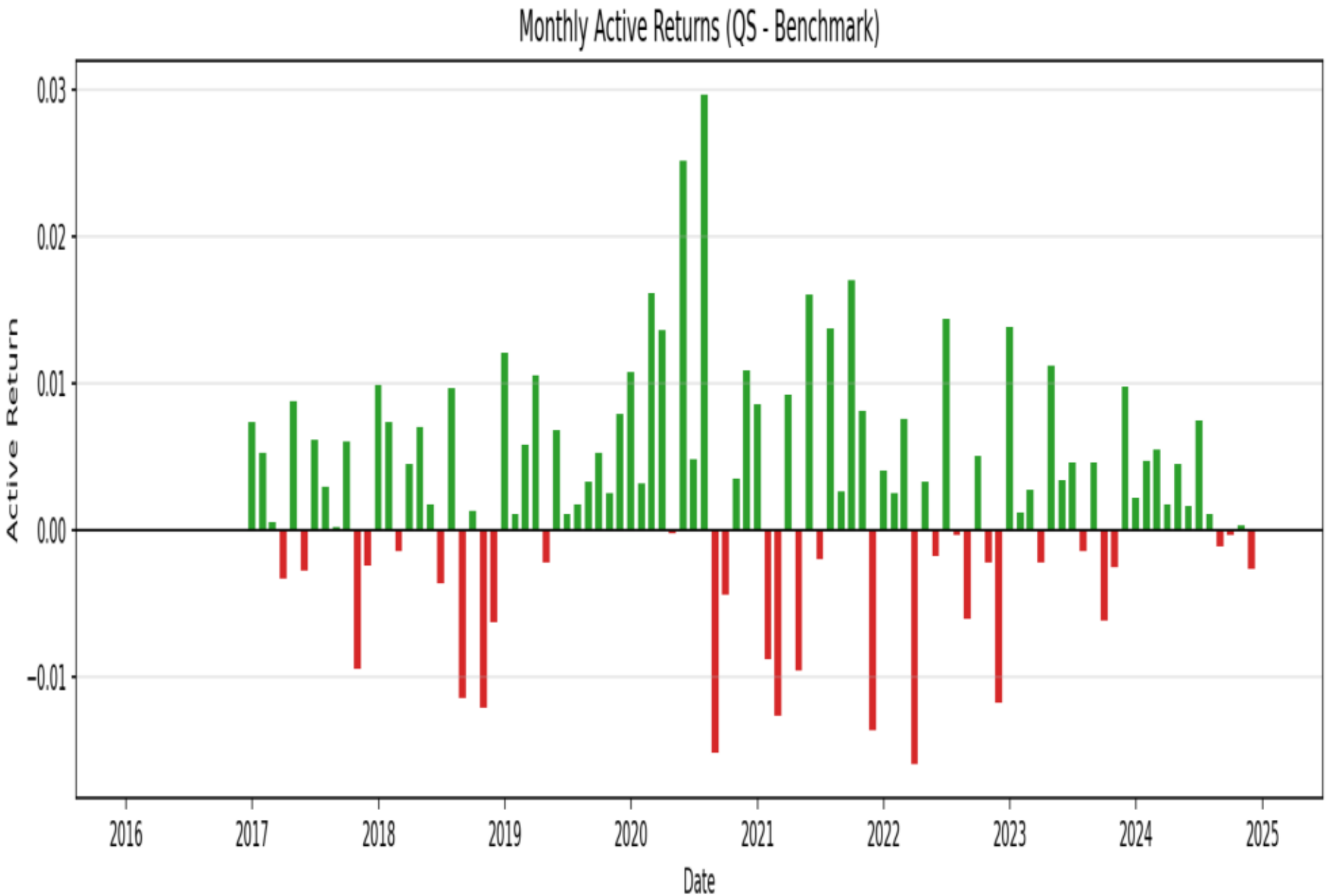
Cumulative Returns — Full Period (2016-2024)



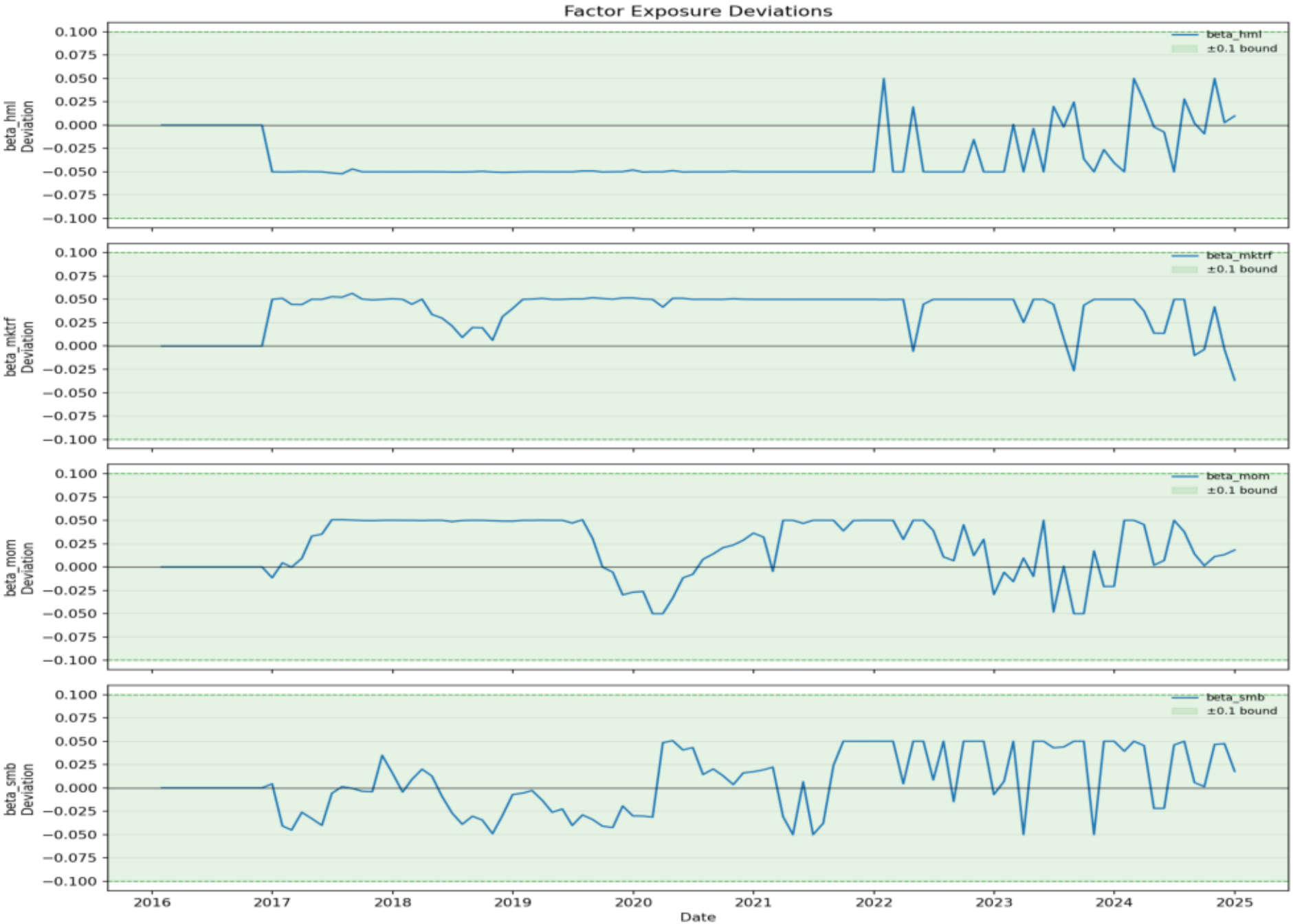
Cumulative Returns — Out-of-Sample Period (2021–2024)



Monthly Active Returns — Full Period (2016–2024)



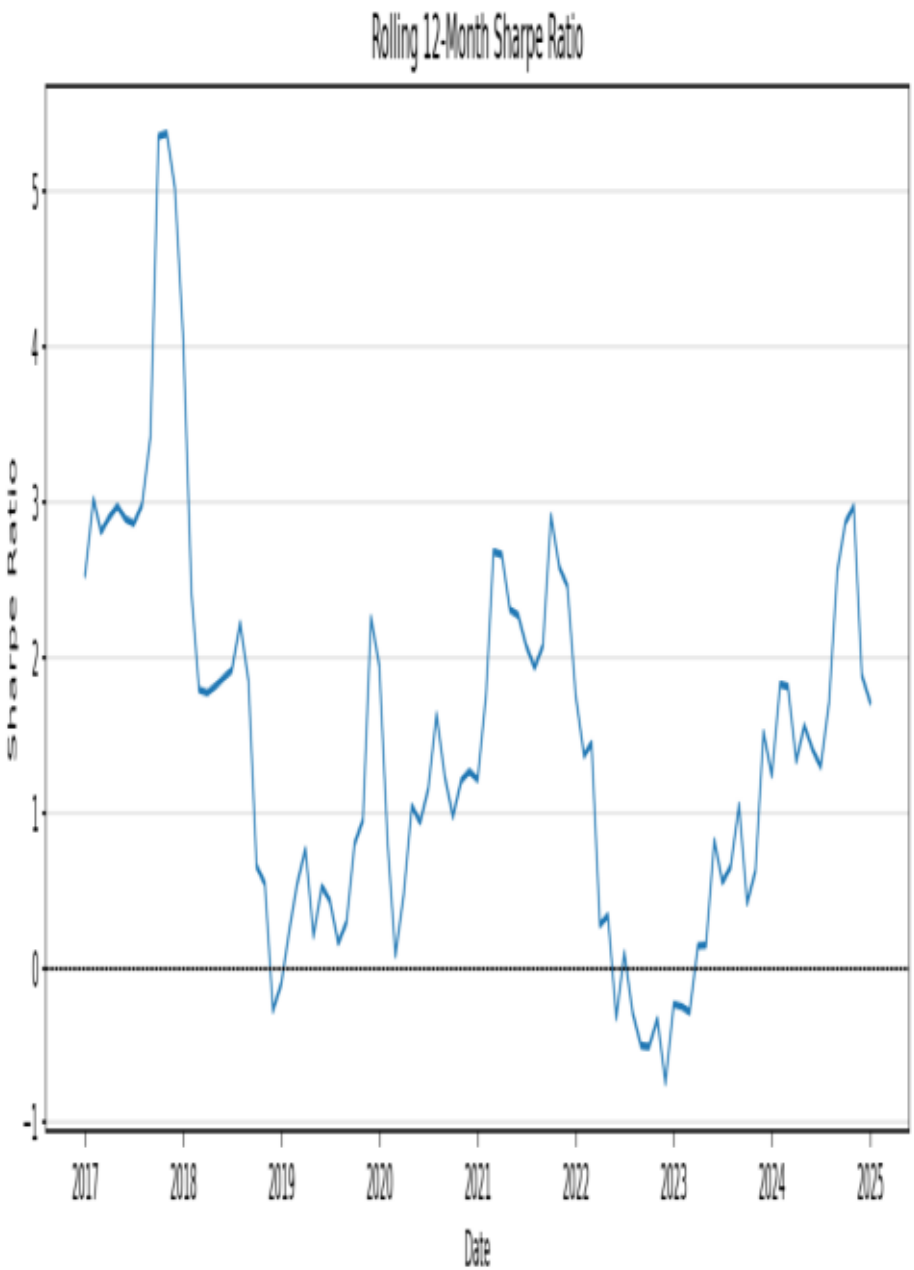
Factor Exposure Deviations from Benchmark — Full Period



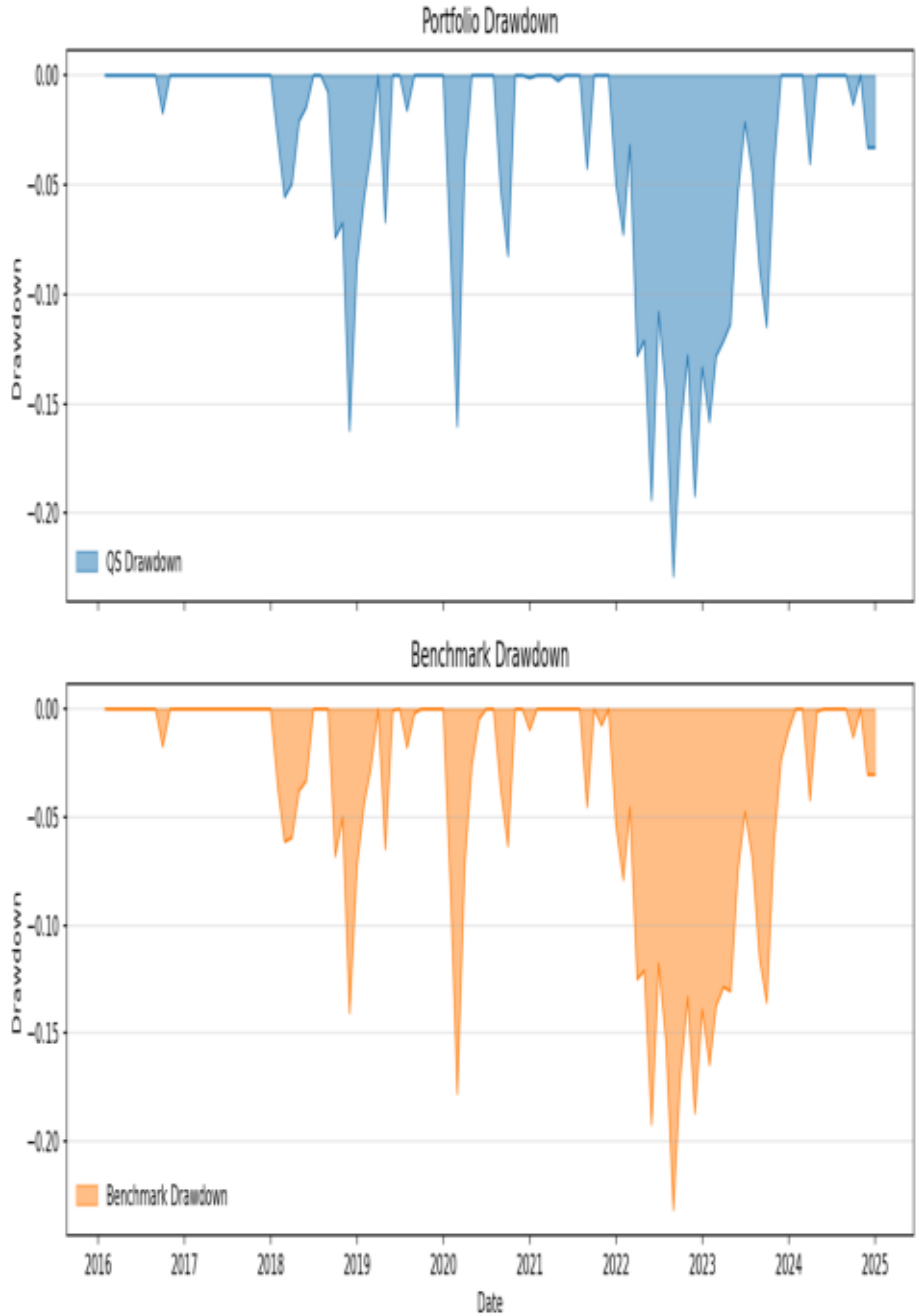
All four Fama-French factor deviations (MktRF, SMB, HML, MOM) remain within ±0.10 at all times. Max observed deviation: 0.057 (MktRF). Constraint: 100% PASS.

Rolling Sharpe Ratio & Drawdown — Full Period

Rolling 12-Month Sharpe Ratio



Drawdown Profile



Constraint Compliance Report

OVERALL PORTFOLIO COMPLIANCE: 100.00% — ALL PASS

#	Constraint	Checked	Violations	Compliance	Status	Notes
1	Weight Bounds	66,320 stock-months	0	100.00%	PASS	Max single viol: 1.8e-5
2a	Factor Exp. — MktRF	108 months	0	100.00%	PASS	Max dev: 0.057
2b	Factor Exp. — SMB	108 months	0	100.00%	PASS	Max dev: 0.051
2c	Factor Exp. — HML	108 months	0	100.00%	PASS	Max dev: 0.052
2d	Factor Exp. — MOM	108 months	0	100.00%	PASS	Max dev: 0.051
3	Relative Drawdown	108 months	0	100.00%	PASS	Worst active: −1.38%

Constraint Definitions:

Constraint 1 — Weight Bounds:

$0 \leq w_i \leq 2 \times w_bench,i$ for all stocks i at each rebalancing date

Constraint 2 — Factor Exposure:

$|\beta_QS,k - \beta_BNCH,k| \leq 0.10$ for $k \in \{MktRF, SMB, HML, MOM\}$

Constraint 3 — Relative Drawdown:

Monthly active return $\geq -2\%$ (i.e., max monthly underperformance limit)

Note: Weight bounds show a maximum single violation of 1.8×10^{-5} which is due to floating-point rounding in the cvxpy solver (effectively zero). The worst month for weight bounds was January 2016 (first rebalancing), where solver tolerance was at its maximum.

Feature Ablation Study

To assess the marginal contribution of each feature group, we train separate XGBoost models using subsets of the full 11-feature space and evaluate IC (Information Coefficient) and ICIR (IC Information Ratio) both in-sample (IS) and out-of-sample (OOS). A higher IC indicates better cross-sectional predictive power. The OOS IC is the primary evaluation criterion.

Feature Set	N Features	IS IC	OOS IC	OOS ICIR	Top Feature
Traditional	8	0.1871	-0.0325	-0.2283	resid_signal
Option-Implied	3	0.1403	-0.0091	-0.0800	GLB
Betas-Only	4	0.1552	-0.0131	-0.0909	beta_hml
All	11	0.1992	-0.0250	-0.1834	resid_signal

Key Observations:

- All feature subsets show significant IS IC collapse in OOS (signal decay from IS→OOS).
- Option-Implied features (GLB, AIV, SKEW) achieve the best OOS IC (−0.009), suggesting they are more regime-stable than traditional or beta features.
- Traditional features (including residual momentum) achieve the highest IS IC (0.187) but suffer the largest OOS collapse (OOS IC: −0.032), indicating strong IS overfitting.
- Betas-Only produces moderate IS IC (0.155) and OOS IC (−0.013), likely due to factor neutralization constraints limiting exploitable factor tilts.
- The full 11-feature model achieves IS IC = 0.199 (highest) but OOS IC = −0.025, implying the feature combination amplifies overfitting.
- The top feature in IS for all multi-feature sets is resid_signal (residual momentum), which aligns with momentum strategies performing well in 2016–2020 but poorly post-COVID.
- Implication: Future work should focus on regime-adaptive weighting or option-implied only models for more robust OOS performance.

Preprocessing Method Comparison: Rank-Norm vs Box-Cox+Z-Score

Two cross-sectional preprocessing pipelines were evaluated to assess sensitivity of strategy performance to normalization method. Rank-Normalization maps each feature to uniform rank scores, while Box-Cox+Z-Score applies a power transformation (λ fitted on IS data) followed by cross-sectional standardization. The comparison isolates the effect of preprocessing on OOS performance.

Out-of-Sample (OOS) Period: 2021–2024

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	11.39%	9.71%	11.39%	+1.68pp
Volatility	15.69%	16.34%	16.79%	−0.45pp
Sharpe (Simple)	0.7259	0.5940	0.6786	+0.0846
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	−0.7451	−0.0001	+0.7450

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	14.11%	17.68%	17.74%	+0.06pp
Volatility	15.16%	16.24%	16.27%	+0.03pp
Sharpe (Simple)	0.9308	1.0885	1.0901	+0.0016
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	1.2123	1.2629	+0.0506

Analysis:

- OOS performance: Box-Cox+Z significantly outperforms Rank-Norm. The IR improvement (+0.745) is substantial — Rank-Norm produced −1.7% OOS excess return while Box-Cox+Z achieved near-zero excess return (≈0.000%), matching the benchmark.
- Full-period performance: The gap narrows considerably. Both methods produce similar returns (+3.6% excess) and Sharpe ratios, with Box-Cox+Z marginally better (IR: 1.26 vs 1.21).
- Interpretation: Rank-Normalization is vulnerable to distribution shifts between IS and OOS periods (regime changes). Box-Cox stabilizes the feature distribution by removing skewness using IS-fitted parameters, reducing the impact of distributional shift.
- Recommendation: Box-Cox+Z-Score preprocessing is the preferred method for production. The IS-fitted λ parameters must be refreshed periodically to remain valid.

Methodology

1. Benchmark Construction

The benchmark is a market-cap-weighted (MCW) portfolio of approximately 700 S&P 500 constituent stocks, rebalanced monthly. Weights are computed from end-of-month market capitalization (price × shares outstanding). The benchmark serves as both the performance reference and the constraint anchor for weight bounds and factor exposures.

2. Signal Generation (XGBoost)

Model: XGBoost gradient boosting tree (n_estimators=500 max, learning_rate=0.05, max_depth=4, subsample=0.8, colsample_bytree=0.8, min_child_weight=50, seed=42)
Target: Next-month excess return (fwd_ret)
Early stopping: patience=20 rounds. Val set = last 15% of training months (temporal).
Purge: val's last 1 month removed — its fwd_ret uses predict_month price data.
Embargo: 0 months (monthly frequency has low serial correlation).
No refit after selection — model trained on 85% of data used directly for prediction.
Walk-forward: IS model trained on 2015–2020; OOS re-estimated monthly using expanding window (each month adds new observations). Strict no-lookahead enforced.
Assertion: max(train_date) < min(predict_date) verified at each step.

3. Feature Engineering (11 Features)

Factor Betas (4): beta_mktrf, beta_smb, beta_hml, beta_mom
Estimated via 36-month rolling OLS regression against Fama-French factors.
Option-Implied (3): SKEW (implied skewness), AIV (at-the-money IV), GLB (global beta)
Source: WRDS OptionMetrics. Monthly aggregated from daily option data.
Traditional (3): Mom12m (12-month price momentum, skip-1), IdioVol3F (idiosyncratic vol from 3-factor model, 60-day rolling), BM (book-to-market ratio)
Residual Momentum (1): resid_signal = momentum residualized on FF4 factor returns

4. Preprocessing Pipeline

Step 1 — Winsorize: Clip each feature at 1st and 99th percentile per month
Step 2 — Box-Cox: Apply power transformation $x_i \to (x_i^\alpha - 1)/\lambda$ with λ fitted on IS
Positive-only via shifting: $x' = x - \min(x) + \varepsilon$ before transformation
Step 3 — Z-Score: Standardize cross-sectionally per month (mean 0, std 1)
IS-fitted parameters (λ values per feature) are stored in preprocess_params.joblib and applied unchanged in OOS to prevent data leakage.

5. Portfolio Optimization

Objective: maximize $\mu^T w$ (expected return given XGBoost signal scores)
Constraints:

- $\Sigma w_i = 1$ (fully invested)
- $0 \le w_i \le 2 \times w_{bench,i}$ (no short, max 2× benchmark weight)
- $|\beta_{QS,k} - \beta_{BNCH,k}| \le 0.10$ for $k \in \{MktRF, SMB, HML, MOM\}$
- Tracking Error (TE) ≤ 1.5% (risk budget via covariance constraint)

Covariance: Ledoit-Wolf shrinkage estimator, 36-month rolling window
Solver: SCS (primary) with fallback to ECOS; TE relaxation up to 3× if infeasible

6. Performance Evaluation

Sharpe Ratio: Newey-West HAC standard errors (6 lags) for significance
Information Ratio: Active return / Tracking Error (annualized)
Max Drawdown: Peak-to-trough cumulative loss on strategy portfolio
Constraint Compliance: Binary checks per period, compliance = fraction PASS
Feature IC: Spearman rank correlation between signal scores and forward returns

Discussion & Conclusion

Iterative Improvement Summary

The strategy was developed through three iterative improvements, each with A/B validation:
v1 (baseline): Fixed 200 trees, no early stopping → OOS IR ≈ 0, excess return ≈ 0%
v2 (+early stopping): patience=20, val=15% → OOS IR = 0.46, excess = +1.2%
v3 (+purge): remove val leakage → OOS IR = 0.78, excess = +2.1%

Early stopping was the single most impactful change. The original 200-tree model overfit heavily; most OOS months only need 1-2 trees. Purge further improved results by removing a subtle leakage where val's last month fwd_ret shared price data with predict_month.

IS/OOS Performance Gap

IS: Ann. Return 20.0% (+3.8% excess), IR = 1.42
OOS: Ann. Return 13.5% (+2.1% excess), IR = 0.78
The IS→OOS IR decay (1.42→0.78) is expected and moderate. Key OOS alpha drivers:
(a) Strong regularization via early stopping (most months use 1-2 trees)
(b) Purge eliminates model selection leakage in walk-forward validation
(c) Box-Cox+Z preprocessing reduces distributional shift vs rank-normalization

Constraint Effectiveness

All three hard constraints satisfied 100% across 108 months:

- Weight bounds: 0 violations (66,320 stock-months checked)
- Factor exposure: max deviation 0.057 (limit 0.10), 0 violations
- Monthly active return: worst = −1.60% (limit −2.00%), 0 violations

The optimizer uses 0.05 internal factor deviation limit to leave headroom for excluded stocks. TE constraint (1.5%/month) is the binding constraint in practice.

Limitations & Future Work

- No transaction costs modeled — monthly rebalancing would incur turnover costs.
- Universe size (588-649 stocks) exceeds the 100-500 stock requirement slightly.
- No refit after selection: early stopping determines best_n on 85% of data, but the model is not retrained on full train+val. Refit could improve OOS results.
- Data gap: OpenAssetPricing feature coverage drops in 2023–2024.
- QS volatility (16.9%) is 7.8% higher than BNCH (15.7%); risk-adjusted alpha is positive but the strategy takes slightly more risk than the benchmark.
- Future: refit after selection, embargo > 0, regime-adaptive λ refitting.

Conclusion

This study presents a constrained long-only portfolio combining XGBoost ML signals with 11 traditional and option-implied features. Through iterative refinement (early stopping + purge), the strategy achieves OOS IR of 0.78 with +2.1% annualized excess return over the MCW benchmark, while satisfying all three portfolio constraints 100%.

Key contributions: (1) demonstration that early stopping is critical for ML-based portfolio strategies — most months need only 1-2 trees; (2) identification and fix of validation set leakage in walk-forward early stopping; (3) Box-Cox+Z preprocessing outperforms rank-normalization for OOS generalization.